Scope 3 Emissions Estimation from Procurement Text: Project Plan

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1. Refined problem and task

Input: one purchase record: free-text description, merchant/recipient, optional structured field (P-card MCC or federal PSC), dollar amount, date. **Output:** (i) a NAICS-6 (2022) code, (ii) emissions = spend in 2022 USD × EPA v1.3 factor (kgCO2e/USD, purchaser price), (iii) a confidence score and an **accept / abstain** decision, with abstained items ranked for human review by *expected emissions error*. Two changes from the proposal: abstention is now a **core** deliverable, and evaluation mixes clean federal contracts with hand-labeled messy purchase-card data. Most of the effort goes into retrieval quality, calibration and error analysis, not prompt wording.

2. How the plan answers the proposal feedback

Feedback	What we found or will do
1. Can one definition map to several codes?	**Checked on the Census files.** Every 6-digit definition is unique (1,012 codes), and none of the 20,398 index phrases maps to more than one code. The many-to-many problem is on the *query* side: 1,076 head nouns span several codes (*pipe, bags, boxes, furniture*: 10 codes each), and 4,601 cross-references ("…except X, see 3329xx") link neighboring codes. We index each index phrase as its own chunk, **aggregate scores by code**, and show the LLM cross-refs as "not this code" hints. In the other direction, EPA factors collapse 1,016 codes into **281 distinct factor values**, so we also report *factor-equivalent* accuracy.
2. Similarity between messy text and formal definitions is low; what is retrieval quality?	We expect low, compressed cosine scores, so **we will never threshold on raw similarity or use it as confidence**. Experiment E1 measures the score distribution (gold vs. top non-gold) per split and chunk type, and judges retrieval by **recall@k** (k = 1–50). P-card text has a median of **2 words** vs. 47 for a definition, so we add short index-phrase chunks, BM25 hybrid search, LLM query rewriting and a cross-encoder reranker (E2).
3. How is the dollar amount used?	(a) Emissions after deflating to 2022 USD. (b) **Review priority** = spend × P(wrong) × factor spread across top candidates (factors range 0.03–3.92 kgCO2e/$, 135×). (c) A tested input feature (a $40 vs. $4M "software" line is a license vs. custom development). (d) Data handling: 1,281 negative P-card rows (refunds), and awards up to $12.6B that would dominate a summed error.
4. Gov data is too clean	Test set = **300 gov + 200 hand-labeled P-card** rows, every metric reported per split. Calibration uses a separate 100-row messy dev split, so the abstention threshold is not fit only on clean data.

3. Simplest credible baselines

- **B0 Retrieval only:** top-1 code from BM25 / dense search over NAICS text, with no LLM. This is the floor the LLM must beat.
- **B1 Supervised:** TF-IDF + logistic regression trained on ~6,400 USAspending (description, NAICS) pairs. This tests whether generative AI is needed at all.
- **B2 Zero-shot LLM:** description → NAICS-6 directly (the proposal's baseline). **B3 MCC lookup** (messy only): hand map of the most common MCC categories to NAICS.

4. Proposed architecture

`record → normalize (strip contract boilerplate, expand abbreviations) → [query rewrite] → hybrid retrieval over chunks {title, definition, index phrase, cross-ref} → aggregate by code → rerank top-50 → LLM selects among top-k with retrieved few-shot examples → confidence signals → accept / abstain → 2022→2017 concordance → emissions`

- **Corpus** (`src/naics_corpus.py`): ~1,012 codes × (title + definition + ~13 index phrases) plus cross-refs. **Retriever** (`src/retrieval.py`): e5/bge with query/passage prefixes + BM25, fused with reciprocal rank fusion. MCC/PSC used as a soft metadata boost (the metadata-filtered RAG pattern from week 7), tested as an ablation.
- **Selector** (`src/classify.py`): structured JSON output {code, rationale, confidence}; few-shot examples are nearest neighbors from the gov pool. **Confidence** (`src/confidence.py`): self-consistency over 5 samples, code log-probability, top-1/top-2 retrieval margin, verbalized confidence, and a logistic-regression combination fit on dev data. **Emissions** (`src/emissions.py`): map 2022 codes to 2017 factors via the concordance; 1→many splits use the mean factor and are flagged.
- Every stage logs its inputs and outputs per item (candidates, gold rank, samples, tokens, latency), so each error can be traced to the stage that caused it.

5. Evaluation data and test cases

Split	Source	n	Labels / purpose
gov_pool	USAspending FY2024 (have 6,900; 23 sectors × 300)	~6,400	Few-shot pool, B1 training, gov calibration dev
gov_test	USAspending, sector-stratified, held out	300	Award NAICS; **50 audited blind** by the team (officers sometimes choose codes for size-standard reasons)
messy_dev / messy_test	OK P-card, July 2024 (34,138 rows), stratified by MCC	100 / 200	**Hand-labeled by 2 annotators**, adjudicated by the third; Cohen's κ at 2/4/6 digits; annotator confidence recorded
challenge	Curated from both sources	~40	Unit-style cases for the edge cases in §8; run on every change

6. Success metrics and methodology

- **Primary:** emissions error, $\Sigma|kgCO2e_{pred} - kgCO2e_{true}| / \Sigma\ kgCO2e_{true}$, plus a per-item median so a few $1B awards do not decide the result. **Secondary:** accuracy at 2/4/6 digits, factor-equivalent accuracy, retrieval recall@k, and cost and latency per item.
- **Abstention (core):** AUROC of confidence for predicting a wrong answer; risk–coverage curve and AURC; ECE with a reliability diagram; emissions error at 70/80/90% coverage. **Does confidence track real errors?** On the human-labeled messy set we also report the Spearman correlation between model confidence and annotator agreement, and the share of human-marked UNKNOWN rows the system abstains on.
- All comparisons use 95% bootstrap CIs and paired bootstrap tests on the same items. Success means beating B1/B2 on emissions error on *both* splits, and AUROC ≥ 0.75 on messy_test.

7. Outcome vs. trajectory evaluation

Outcome: final code, emissions and the accept/abstain decision (§6). **Trajectory:** the pipeline is a fixed chain, so we score each stage, following the trajectory logging from the week 8 agent assignment. Was the gold code retrieved (gold rank)? Did query rewriting move it up or down? Given that it was in the top-k, did the LLM pick it? Did confidence flag the miss? This splits every error into **retrieval miss / selection error / label noise / missed abstention**, with each category reported by count and by share of emissions error. If the stretch agentic variant is built (the LLM can re-query or walk the NAICS hierarchy), we also score tool-call count and redundant steps.

8. Important edge cases

Case	Evidence in our data	Handling
No product signal	39% of P-card rows are GENERAL PURCHASE / PAYMENT ON ACCOUNT	Fall back to merchant + MCC; expected to abstain; label UNKNOWN if unknowable
Retailer ≠ producer	18% of P-card rows are Amazon; Walmart and Sam's Club also common	Label the producing industry; the MCC is often misleading
Multi-item / truncated	"PATIO RENTAL…|MEAL VOUCHERS", "Hea PCE"	Dominant-dollar item; rewrite step expands abbreviations
Refunds, extreme spend	1,281 negative rows; awards up to $12.6B	Net refunds against purchases; report per-item and dollar-weighted error
Contract boilerplate	"CONTRACT W58RGZ-17-D-0055, TASK ORDER…"; 11 near-empty	Strip IDs; empty text must abstain
Goods vs. services; NAICS 2017↔2022	"software", "HVAC"; mixed code versions in labels	Amount as feature; concordance with split flags

9. Planned experiments

- **E1 Similarity:** cosine distributions for gold vs. hardest negative, per split and chunk type. **E2 Retrieval ablations** (recall@k): dense → asymmetric prefixes → + index chunks → + BM25 hybrid → + rewrite → + rerank → + MCC/PSC boost.
- **E3 Classifier stages:** B0–B3 vs. RAG vs. RAG + few-shot; $k \in \{5, 10, 20\}$; with and without amount and structured field. **E4 Confidence:** each signal alone vs. combined; calibrated on gov-dev vs. messy-dev. **E5 Review queue:** abstain by P(wrong) vs. by expected emissions error (emissions error recovered per item reviewed). **E6 (stretch):** small vs. large model.

10. Core and stretch goals

Core: NAICS corpus and overlap report; hybrid retriever with E1/E2; B0–B3 and the RAG + few-shot classifier; **confidence-based abstention with AUROC, AURC and ECE**; 300 double-labeled P-card rows with κ; stage-attributed error analysis weighted by emissions. **Stretch:** contrastively fine-tuned embedder on gov_pool pairs; LoRA-tuned small selector model (week 9); hierarchical (2→4→6 digit) classification; agentic re-query; a small review-queue demo.

11. Team ownership (proposed)

Member	Owns	Also
Rishik Kolli	Corpus, overlap report, retriever, E1/E2, repo and data pipeline	Labels messy rows; adjudicates Bineet/Simranjeet disagreements
Simranjeet Singh	LLM selector, few-shot, query rewrite, emissions module, B1–B3, E3/E6	Labels messy rows; gov audit (25)
Bineet Singh Chadha	Confidence signals, abstention, eval harness and bootstrap, E4/E5	Labels messy rows; gov audit (25)

Labeling load: 300 messy rows × 2 labels = 600 labels, about 200 per person, plus the 50-row gov audit. Everyone writes the error-analysis section for their own stage.

12. Next steps

- **Week 1:** freeze splits (`scripts/build_splits.py`); draw the 300-row P-card sample; label 30 rows together to align on guidelines; build the corpus.
- **Week 2:** E1 + E2 retrieval ablations; B0–B2 on gov_test; first labeling round with κ. **Week 3:** RAG + few-shot (E3); finish labeling; gov audit. **Week 4:** confidence and abstention (E4/E5). **Week 5:** error analysis, stretch goals, report.